TODO: Jag har försökt lägga fram kärnan i vad jag tror kommer att bli uppsatsen, men många delar är ännu inte färdiga. Jag har försökt lämna färgade kommentarer här och var så att du kan se hur jag har tänkt samt vilka frågeställningar jag har.
Utöver det är jag inte helt säker på mina val av felmodeller (convolution) och stickprovsstorlekar. Jag har skrivit min kod i R som en pipeline, så dokumentet kan enkelt uppdatera befintliga grafer och plottar om du tycker att jag bör använda andra värden.
Jag är också osäker på hur mycket av materialet som bör tas med i uppsatsen i dess nuvarande form. I dagsläget har jag 64 olika simuleringar, vilket är en kombination av felmodell, stickprovsstorlek och dekonvolutionsmetod.

To be determined)

Emilio Otero Johansson

April 20, 2026

Abstract

TODO: Will be done at the end

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1 Introduktion

TODO:

- **Background:** Introduce existing age determination methods, in particular recent developments Nakamura et al. 2023. Discuss why accurate age determination is important for understanding population dynamics and improving population forecasts.
- **Problem statement:** Measurement methods introduce inherent error, causing observed age distributions to deviate from the true underlying distribution. If uncorrected, this can lead to biased population estimates and unreliable forecasts. This thesis addresses this issue by accounting for such errors.
- **Convolution:** Provide a brief explanation of convolution and its role in modeling measurement processes.
- **Scope of this thesis:** Conduct a simulation study to evaluate a deconvolution method based on the EM algorithm. The method is applied to a hypothetical age distribution generated using a Leslie matrix, together with theoretical error models based on Norwegian Institute for Nature Research (NINA) n.d., across varying sample sizes. Performance is assessed using total variation distance and a forecast simulation.
- **Optional?:** use the error model of Nakamura et al. 2023 and asses it with above given methods.

2 Methodology

2.1 Deconvolution

Measurement error in statistical models can be described as the discrepancy between a latent (true) variable and its observed counterpart. A common way to formalize this is through a convolution structure, where the observed variable is a noisy transformation of the true signal.

$$Y = Z + \epsilon.$$

Here, Z denotes the latent random variable, ϵ denotes measurement error, and Y denotes the observed random variable. The distribution of Y is then the convolution of the distributions of Z and ϵ.

This formulation extends naturally to multivariate settings, where both the latent and observed quantities are represented as random vectors.

$$\mathbf{Y} = \mathbf{Z} + \epsilon.$$

In many applications, including the present one, the variable of interest is discrete. In such cases, the additive error formulation is replaced by a discrete measurement error mechanism, which will be specified in the following sections.

In a deconvolution problem, the goal is to recover the distribution of the latent variable based on observations of the corresponding noisy measurements and under the assumption that the distribution of the error term is fully known. Deconvolution problems are typically ill-posed in the sense that small perturbations in the observed data can lead to large changes in the estimated distribution (e.g. due to sampling variation).

Latent distribution In this study, the true age structure of the bear population is represented by the parameter vector

$$\boldsymbol{\pi} = (\pi_1, \pi_2, \ldots, \pi_k),$$

where π_j denotes the probability that an individual belongs to age class j. In particular, π_1 corresponds to the age group 0–1 years, π_2 to 1–2 years, and so on. The discretization of age into fixed classes is also consistent with population structure is often analyzed in biologically meaningful age intervals, where demographic processes such as survival and reproduction are age-class dependent as seen in Norwegian Institute for Nature Research (NINA) n.d.

Let n denote a number of independent observations drawn from this distribution. The random vector $\boldsymbol{Z} = (Z_1, Z_2, \ldots, Z_k)$ represents the number of individuals in each age class for a sample of size n. We assume that

$$\boldsymbol{Z} \sim \text{Multinomial}(n, \boldsymbol{\pi}).$$

Measurement error As mentioned in section 2.1 the measurement error is assumed to be fully known. In this simulation study, we assume a normal distribution for the error term as a simple and tractable way to control the variability in the observations and to introduce systematic measurement error. While this choice may not exactly reflect the error structure described in Nakamura et al. 2023, the aim of this thesis is to investigate the recovery of the latent distribution under a simplified but comparable error setting.

Additionally, we assume that the measurement error depends on the true age class j. Specifically, we model

$$e \mid j \sim \mathcal{N}(f(j), g(j)).$$

where both the mean and variance of the error may vary across age classes. Applying the measurement error as an additive effect on the latent random

vector $\mathbf{Z}$ would yield a random vector of continuous scalar values $\mathbf{Y}$ thus violating the discrete setting of our model. We therefore bin the probability mass of our error model $\epsilon \mid j$ in to the age classes j. let:

The probabilities p_{ij} denote the probability of misclassifying a bear of true age j with age i which is obtained by discretizing the conditional distribution of $\epsilon \mid j$ over the predefined age intervals such that:

$$\sum_{i=1}^{k} p_{ij} = 1$$

This leads to to the construction of a square missclassification matrix M of dimension k where M_{ij} corresponds to p_{ij}. That is, row represents the observed age class and column true age class.

Observed data Given the misclassification mechanism described above, each individual from true age class j is observed in age class i with probability p_{ij}. Consequently, the observed counts $\mathbf{Y} = (Y_1, \ldots, Y_k)$ follow a multinomial distribution with probabilities given by the transformed latent distribution:

$$\mathbf{Y} \sim \text{Multinomial}(n, M\boldsymbol{\pi}).$$

Naive estimator The objective of this thesis is to investigate how well the true age distribution $\boldsymbol{\pi}$ can be recovered from misclassified observations. In the absence of measurement error, a natural estimator of $\boldsymbol{\pi}$ is given by the empirical proportions

$$\hat{\boldsymbol{\pi}} = \frac{\mathbf{Z}}{n},$$

which satisfies $\mathbb{E}[\hat{\boldsymbol{\pi}}] = \boldsymbol{\pi}$.

However, in the presence of measurement error, we observe $\mathbf{Y}$ instead of $\mathbf{Z}$, and the naive estimator $\mathbf{Y}/n$ is generally biased for $\boldsymbol{\pi}$. This motivates the deconvolution approach considered in this thesis.

2.2 Matrix population models

To obtain a suitable idea of the population age distribution over time we use a population matrix model Caswell 2001 which we will describe shortly in this section. The age class $i = 1, 2, ...m$ is the discretization of the age z to age-class i with $i - 1 \leq z \leq i$. The number of individuals for each age class i is then given by $n_1, n_2, ..., n_m$.

Following Caswell 2001, Eq. 2.1 the number of n_1 individuals at time $t + 1$ can be projected with

$$n_1(t + 1) = f_1 n_1(t) + f_2 n_2(t) + ... + f_i n_i(t)$$

Here f_i denotes the reproductive rate per-capita. That is the average number of offspring an individual of age class i has. For $i > 1$ the number of individuals in an age class n_i at time $t+1$ is simply given by their survival probaility $(s_{i-1})(n_{i-1})$ at the previous timestep t. Together we can write the equation in matrix form for all n_i at time $t+1$.

$$\begin{pmatrix} n_1 \\ n_2 \\ \vdots \\ \vdots \\ n_m \end{pmatrix} (t+1) = \begin{pmatrix} f_1 & f_2 & f_3 & \cdots & f_m \\ s_1 & 0 & 0 & \cdots & 0 \\ 0 & s_2 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & s_{m-1} & 0 \end{pmatrix} \begin{pmatrix} n_1 \\ n_2 \\ \vdots \\ \vdots \\ n_m \end{pmatrix} (t)$$

2.3 Computing age distribution from a Leslie matrix

The stable age distribution of a population can be obtained from the Leslie matrix $\mathbf{A}$ by computing its dominant eigenvalue λ_1 and the corresponding right eigenvector $\mathbf{w}$. This eigenvector gives the relative proportions of individuals in each age class at equilibrium, and can be normalized so that the entries sum to 1.

The existence of a real, non-negative eigenvalue λ_1 that is greater than or equal in magnitude to all other eigenvalues λ_i of a non-negative, primitive matrix $\mathbf{A}$ is guaranteed by the Perron-Frobenius theorem Caswell 2001.

If $\mathbf{A}$ is primitive, then as $t \to \infty$, the population vector $\mathbf{n}_t$ satisfies

$$\frac{\mathbf{n}_t}{\sum_i n_{i,t}} \to \mathbf{w},$$

where $\mathbf{w}$ is the normalized right eigenvector corresponding to λ_1. This represents the **stable age distribution** of the population.

Question: Maybe explain here that w is the parameter of a multinomial distribution?

2.4 Matrix inversion & condition number

[Question: Does this need a source reference? ill posed problems, justification to EM]

2.5 Thikanov regularization

Optional: could be interesting to compare with em-alg
Include SVD formulation, derivation of the filter factor $\frac{\sigma_i}{\sigma_i^2+\lambda}$, and discussion of how regularization trades bias for stability in ill-posed inverse problems. Would justify the use of kl-divergence too

2.6 Expectation-Maximization algorithm (EM-algorithm)

TODO: Har inte infört några förändringar sen senaste feedback

The EM algorithm is a general-purpose method for computing maximum likelihood estimates in the presence of incomplete or latent data (McLachlan and Krishnan 2008). Let $\mathbf{Y}$ denote a discrete random vector corresponding to the observed (incomplete) data, and $\mathbf{X}$ a discrete random vector representing the complete data. If $\mathbf{X}$ were fully observable, we could directly compute the complete-data log-likelihood function $\ell_c(\theta)$ based on a realization x.

However, in practice we only observe $y \in \mathcal{Y}$, the sample space of $\mathbf{Y}$. The complete-data sample space $\mathcal{X}$ is related to $\mathcal{Y}$ through a many-to-one mapping, i.e., a surjection from $\mathcal{X}$ to $\mathcal{Y}$, where multiple elements $x \in \mathcal{X}$ correspond to the same observed value y.

The probability mass function of $\mathbf{Y}$ is therefore given by

$$p_Y(y;\theta) = \sum_{x \in \mathcal{X}(y)} p_X(x;\theta), \tag{1}$$

where $\mathcal{X}(y) = \{x \in \mathcal{X} : h(x) = y\}$.

Since the complete data $\mathbf{X}$ is not observable, the complete-data log-likelihood $\ell_c(\theta)$ cannot be evaluated directly. Instead, the EM algorithm considers its conditional expectation given the observed data y and a current parameter estimate $\theta^{(t)}$:

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}\left[\ell_c(\theta) \mid Y = y\right] \tag{2}$$

which corresponds to the Expectation step (E-step) of the EM algorithm. In the Maximization step (M-step), the parameter is updated by maximizing this function:

$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta \mid \theta^{(t)}). \tag{3}$$

The process is then iterated upon until $l(\theta^{t+1}) - l(\theta^t) <= k$ for some arbitrary k value.

Monotonicity . A key property for the algorithm to work is for the likelihood to be monotonically increasing for each iteration. $L(\theta^{t+1}) \geq L(\theta^t))$ which is discussed in Section 3.2 of McLachlan and Krishnan 2008.

2.7 Application of the EM algorithm to the deconvolution problem

Let $X = (Z, Y)$ denote our complete data, that is in our deconvolution problem the observed age Y and the corresponding true age Z. The PMF for X is then:

$$p_X(x = (z_i, y_j); \theta_Z) = p_i M_{ij} \tag{4}$$

where M is the missclasification matrix defined in **??** and for a vector of n independent observations $\mathbf{x} = (z_1, y_1), (z_2, y_2), ..., (z_n, y_n)$ we obtain

$$p_X(\mathbf{x}; \theta_Z) = \prod_i \prod_j (p_i M_{ij})^{n_{ij}} \tag{5}$$

where $n_i j = \sum_{k=1} 1\{X_k = i, Y_k = j\}$ is a random variable with the number of counts that were observed as j given they were i.

The log-likelihood becomes:

$$\ell_c(\theta_Z) = \sum_i \sum_j n_{ij} \log(p_i M_{ij}) = \sum_i \sum_j n_{ij} \log M_{ij} + \sum_i \sum_j n_{ij} \log p_i \tag{6}$$

In accordance with equation (2) we take the conditional expectation on $n_i j$ with an initial parameter θ^0 conditioned on Y

$$\mathbb{E}_{\theta^{(0)}}\big[\ell_c(\theta) \mid Y = j\big] = \mathbf{E}_{\theta^{(0)}}\big[\sum_i \sum_j n_{ij} \log(p_i M_{ij}) \mid Y = j\big] = \sum_i \sum_j \mathbf{E}_{\theta^{(0)}}\big[n_{ij} \mid Y = j\big] \log(p_i M_{ij}) \tag{7}$$

The conditional expectation $E[n_i j | y = j]$ is the expected number of bears of age i missclasified as age j given the total amount of bears observed as age j.

Taking expectation:

$$\mathbb{E}[n_{ij} \mid Y = y] = \sum_{k=1}^{n} \mathbb{E}\big[\mathbf{1}\{Z_k = i, Y_k = j\} \mid Y = y\big] \tag{8}$$

Now conditioning on the observed Y_k:

$$= \sum_{k:Y_k=j} P\big(Z_k = i \mid Y_k = j, \theta^{(0)}\big) \tag{9}$$

Let $n_j = \#\{k : Y_k = j\}$ denote the number of observations with $Y = j$. Then:

$$\mathbb{E}[n_{ij} \mid Y = y] = n_j \cdot P\big(Z = i \mid Y = j, \theta^{(0)}\big) \tag{10}$$

Using Bayes' theorem:

$$P\big(Z = i \mid Y = j, \theta^{(0)}\big) = \frac{p_i^{(0)} M_{ij}}{\sum_{i'} p_{i'}^{(0)} M_{i'j}} \tag{11}$$

Thus:

$$\mathbb{E}[n_{ij} \mid Y = y] = n_j \cdot \frac{p_i^{(0)} M_{ij}}{\sum_{i'} p_{i'}^{(0)} M_{i'j}} \tag{12}$$

Maximization-step

2.8 Evaluation method

Total Varation (TV) To quantify the difference between two probability distributions P and Q over the same finite sample space $\mathcal{X}$, we use the *Total Variation distance (TV)*:

$$\mathrm{TV}(P, Q) = \frac{1}{2} \sum_{x \in \mathcal{X}} |P(x) - Q(x)|.$$

The TV distance ranges from 0 (distributions are identical) to 1 (distributions have no overlap). Intuitively, it represents the maximum fraction of probability mass that must be reallocated to transform P into Q or vice versa. In this thesis, we use TV distance to quantify and distinguish the effects of sampling variability, convolution, and deconvolution on our estimates of the latent distribution (true age distribution).

[TODO: Does this need a source reference?]

KL-divergence **[Optional: seems unnecessary, it would only be useful if Tikhonov reg**

3 Methods / Simulation Setup

3.1 Simulated Age Distributions

A Leslie matrix L of dimension 20×20 was used to model the age-structured population. The first row represents the expected number of offspring produced by individuals in each age class, calculated as the product of three components for age class i:

$$F_i = s_i \cdot p_{\mathrm{coy},i} \cdot \mathrm{litter}_i, \tag{13}$$

where:

- s_i is the survival probability from age i to $i + 1$,
- $p_{\mathrm{coy},i}$ is the probability of conceiving at age i,
- litter_i is the expected litter size at age i.

The survival and reproduction parameters were estimated from published graphs Norwegian Institute for Nature Research (NINA) n.d. The full data vectors used to construct the Leslie matrix are provided in Appendix 6.1.

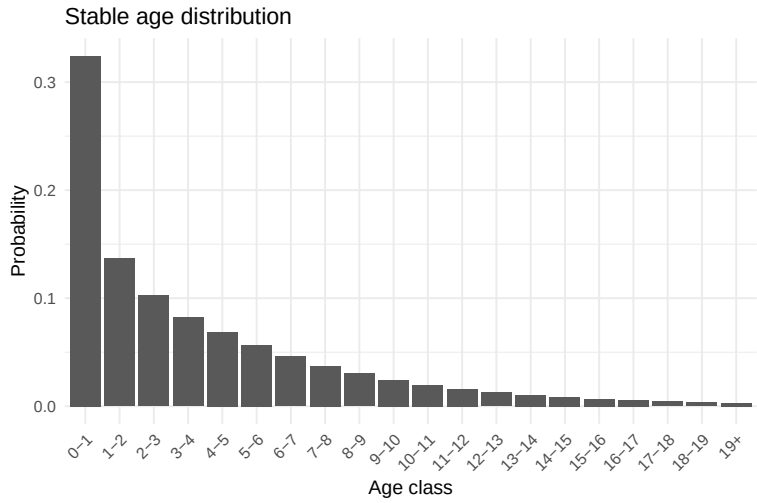


Figure 1: Stable age distribution obtained from the normalized dominant eigenvector of the Leslie matrix (see Section 2.3 for derivation).

3.2 Error terms

To simulate plausible errors in age determination, we considered four different error models. Let Z denote the true age, and e the error term added to Z in the convolution. We model the error as conditionally normal:

$$e \mid Z \sim \mathcal{N}\big(\mu = f(Z), \sigma^2\big),$$

and discretize the resulting values to the integer age bins (1–20) to match the observed age categories. Negative values were excluded, and the remaining probabilities were rescaled by $1/(1 - F(e < 0))$. Values greater than 20 were assigned to the 19+ age class.

Question: I set the mortality for bears over 19+ to 100% because i didnt find data to compute actual mortality or fertility

The following four specifications for $f(Z)$ (mean) and σ (standard deviation) were considered:

Model A: **Model A Unbiased, low variance:** $f(Z) = 0, \quad \sigma = 1$

Model B: **Model B Unbiased, high variance:** $f(Z) = 0, \quad \sigma = 4$

Model C: **Model C Age-dependent bias, low variance:** $f(Z) = \dfrac{4}{Z + 0.5}$, $\sigma = 1$

Model D: **Model D Age-dependent bias, high variance:** $f(Z) = \dfrac{4}{Z + 0.5}$, $\sigma = 4$

TODO: Add reference and explanation for the choice of sigma and mu models. Improve clarity on how the mass redistribution was done with error models. Check for consitency /overlap with section ??.

3.3 Convolution and Sampling Effects

Having constructed the misclassification matrices for the four error models (Section 3.2), we analytically convolve the true age distribution $p_Z(z)$ to obtain the distribution of observed ages Y:

$$p_Y(y) = \sum_z M_{zy}\, p_Z(z),$$

where M is the misclassification matrix corresponding to each error model.

To assess the effects of sampling error, we simulated random samples of size $n = 10, 25, 50, 100, 150, 200$ from the convolved distribution $p_Y(y)$. This allows us to separate variability due to sampling error from information loss caused by the convolution.

The difference between the sampled distribution and the true age distribution was quantified using Total Variation distance.

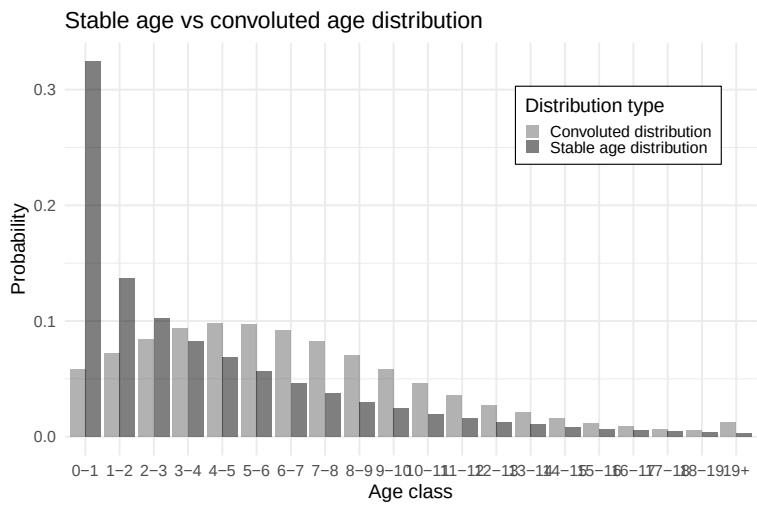


Figure 2: Stable age distribution vs convoluted age distribution using Model D

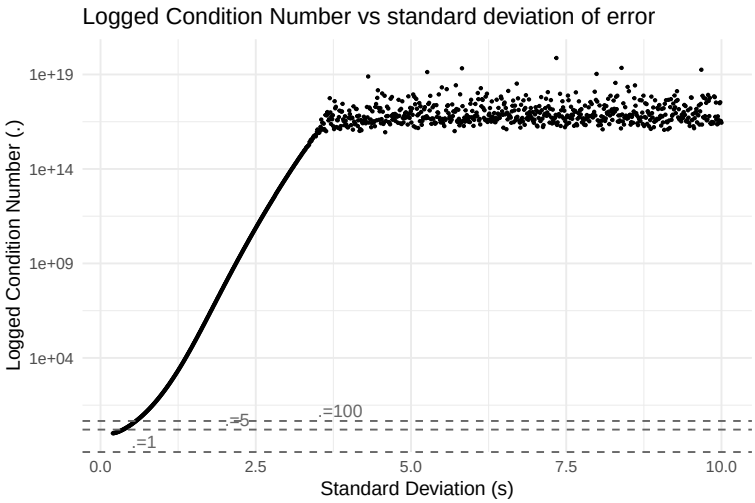
[TODO: Describe how $M^t p$ is the convoluted distribution, refer to section in Theory or write here?]

3.4 Deconvolution Methods

TODO: This section depends if i will have the time or should attempt any other method for deconvolution i leave it as it is for after feedback

Two deconvolution methods were used and compared: a thikanov regularization matrix inversion and the EM algorithm described in Section 2.6. For each error model introduced in Section 3.2. The resulting estimates were evaluated using the KL-divergence, as described in Section 2.8.

Matrix inversion The matrix inversion method inverts the misclassification matrix that can then be used to calculate the recovered distribution, negative estimates are set to 0 and the distribution is then normalized. While straightforward, this approach can become unstable and or ill-conditioned as σ for the error distribution increases as shown in 3.4.



EM-algorithm deconvolution We applied the EM algorithm (see Appendix **??**) using two different initialization: a uniform distribution, representing a neutral/uninformed initial belief about the latent age distribution, and the true age distribution.

The initialization based on the true age distribution was included to assess how a well-informed prior affects the resulting estimate compared to a neutral one. In particular, this allows us to evaluate the sensitivity of the EM algorithm to initialization and to investigate for which sample sizes initialization no longer constitutes a confounding factor in an ideal setting.

The algorithm was iterated until convergence, defined as

$$\ell_c(\theta^{(t+1)}) - \ell_c(\theta^{(t)}) \leq 10^{-4},$$

or until a maximum of 100 iterations was reached.

3.5 Evaluation Against a Random Parameter Baseline

Question:

- I wanted to include this as a sanity check. We may prove that deconvolution is better than convolution in terms of TV however it does not say that we might as well be better off ignoring the data and guessing the parameter randomly.

To assess whether our estimator outperforms a random baseline, we estimate the probability that a randomly sampled parameter from the parameter space yields a smaller total variation (TV) distance to the true distribution than our estimator.

Let $\theta_s^{(i)} \sim \mathcal{U}(\Theta)$ for $i = 1, \ldots, N$ denote independent samples from the parameter space. For each sample, we compute the indicator

$$I^{(i)} = \begin{cases} 1 & \text{if } TV(\theta_s^{(i)}, \theta_{true}) < TV(\hat{\theta}, \theta_{true}), \\ 0 & \text{otherwise.} \end{cases} \tag{14}$$

The probability is then estimated using a Monte Carlo approximation:

$$\hat{P} = \frac{1}{N} \sum_{i=1}^{N} I^{(i)}. \tag{15}$$

This quantity represents the empirical probability that a randomly chosen parameter from the parameter space outperforms the estimator in terms of total variation distance to the true distribution.

4 Results

TODO: text and polish is largely unfinished but the main results to describe are there in plots and tables

4.1 Effect of convolution and sampling

Figure **??** shows how sample size affects the total variation distance (TV distance) across all models. For small sample sizes, sampling noise significantly affects the TV distance. As the sample size increases, the TV distance gradually stabilizes across all models, suggesting that sampling noise plays a diminishing role relative to the structural effects induced by the convolution. Sampling from the true model is included as a theoretical baseline for sampling noise.

For the unbiased Models A and B, most of the variation in TV distance is driven by sampling noise, whereas for the biased Models C and D,

the TV distance is primarily driven by convolution structure. Additionally, Model C, which is characterized by lower sd ($\sigma = 1$) compared to Model D ($\sigma = 4$), exhibits systematically higher total variation distance.

The median and non-parametric 95% upper quantile also exhibit a consistent decreasing pattern with increasing sample size, in line with the results observed for the mean (see Table **??**).

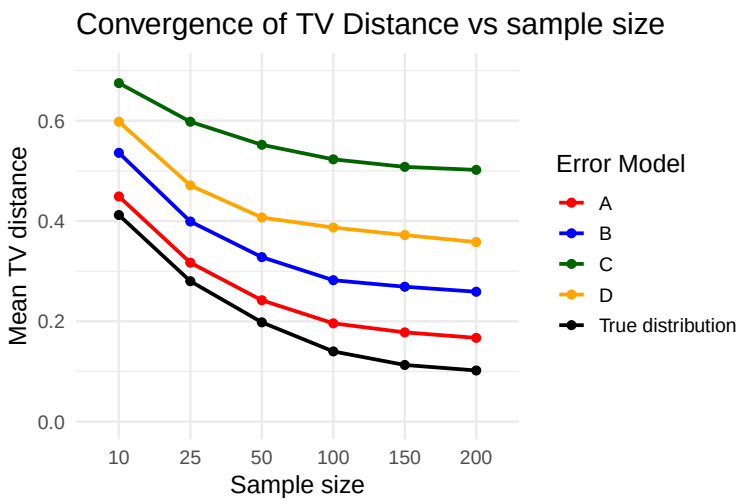


Figure 3: Mean TV distance for convoluted distributions for Model A–D, with 100 replicas per sample size. Total variation distance is computed relative to the true distribution.

4.2 Deconvolution

Figure 4 and Figure **??**

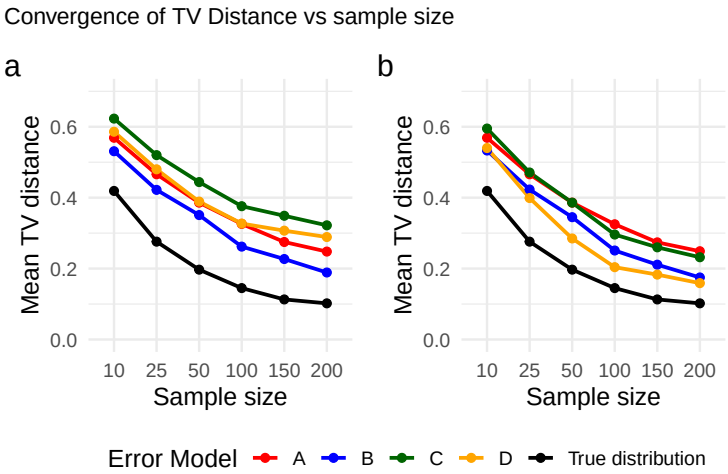


Figure 4: Mean total variation (TV) distance for deconvolved samples based on an uninformed initialization for Models A–D, using 100 replicates per sample size. The TV distance is computed relative to the true distribution. Panel (a) shows the EM algorithm with a uniform (uninformed) initialization, while panel (b) shows results with initialization using the true (informed) distribution.

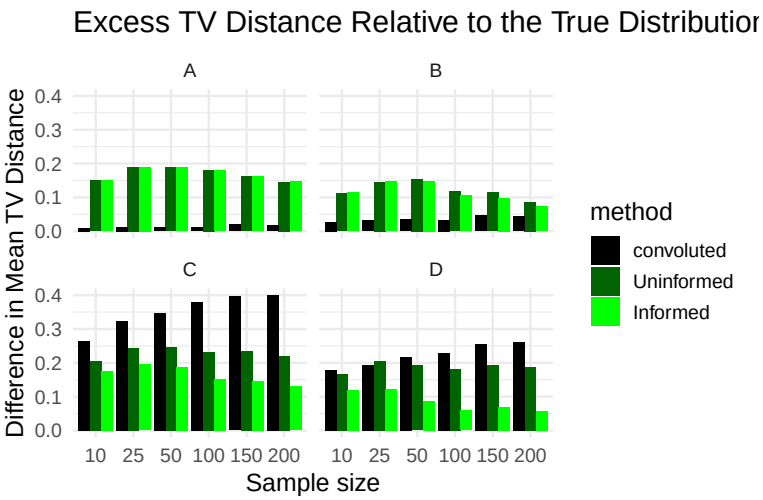


Figure 5: Mean total variation (TV) distance for convolved and deconvolved samples based on an uninformed and informed initialization, using 100 replicates per sample size. The TV distance is computed relative to the true distribution. Panel (A–D) corresponding to error Models A–D.

model	sample_size	comparison	diff	stars
C	200	I-C	-0.262	***
C	150	I-C	-0.247	***
C	100	I-C	-0.224	***
D	200	I-C	-0.190	***
D	150	I-C	-0.185	***
C	200	U-C	-0.175	***
D	100	I-C	-0.173	***
A	25	U-C	0.163	***
A	25	I-C	0.162	***
C	50	I-C	-0.161	***

Table 1: Table displaying the top 10 differences in total variation (TV) distance between convoluted and deconvoluted samples. Statistical significance is assessed using paired t-tests, where * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.

TODO: Plot 6 and Plot 7 tell the main story how sample size, deconvolution vs convoluted samples affect total population forecasts. I didnt have time to describe these properly and what can be observed. TV was used more as a diagonostic tool for the deconvolution but does not say much about weather or not forecast would be good using it. E.g. if most of the TV can be explained my shifts in the probability mass of age classes that have 0 fertility this would not affect the total population forecast the coming year. The plots were made by simulating 100 samples for each combination and taking $L^n(p_{error} - p_{true})$ for $n = 1, 2, ...5$

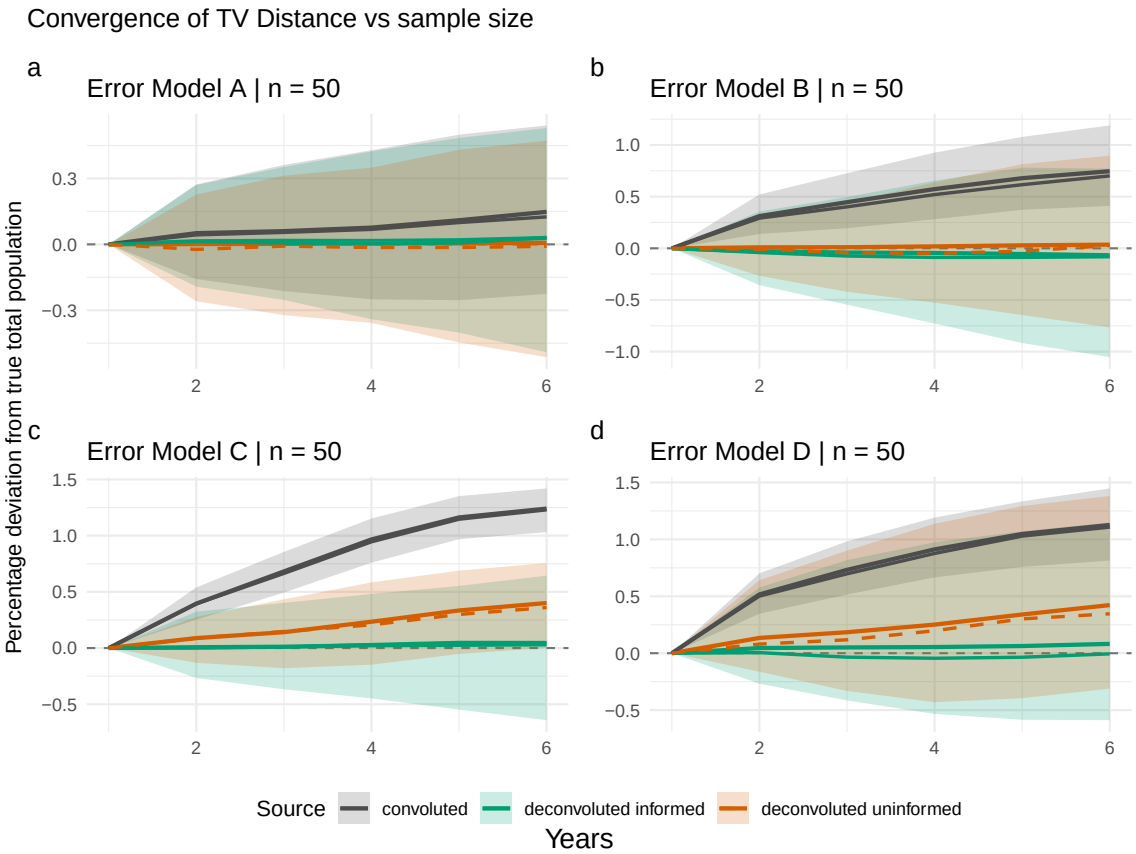

Convergence of TV Distance vs sample size
a
Error Model A | n = 50
b
Error Model B | n = 50
c
Error Model C | n = 50
d
Error Model D | n = 50
Percentage deviation from true total population
Source
convoluted
deconvoluted informed
deconvoluted uninformed
Years

Figure 6: s

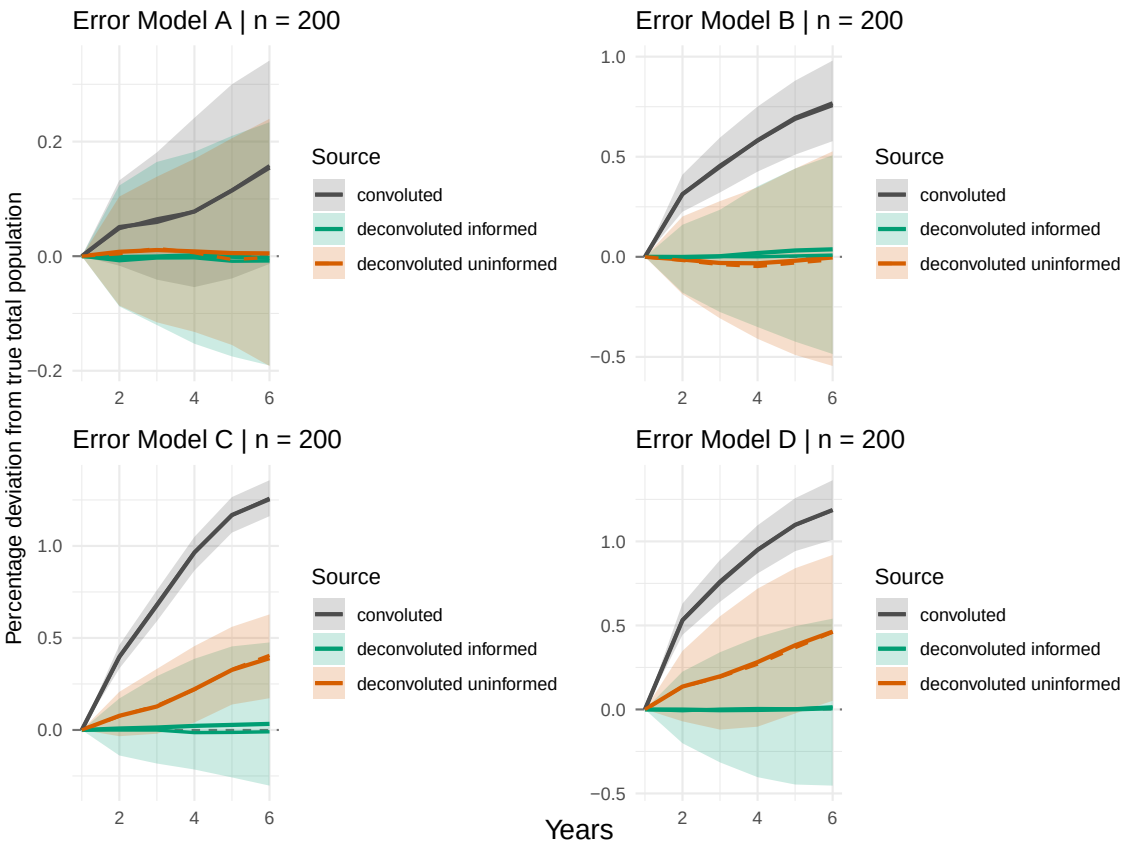


Figure 7: s

5 Analysis / Discussion

TODO:

- Under which conditions the deconvolution method is reliable and appropriate, in terms of total variation (TV), sample size, and error structure.
- Analyze how sampling variability interacts with different error models, and whether it affects deconvolution performance differently across these settings.
- Compare informed vs uninformed deconvolution approaches vs keep convoluted. Assess whether informed initialization leads to improved performance, and how this relates to convergence properties of the expected conditional log-likelihood, including the probability of convergence to different local optima. Suggest as improvement whether multiple random initializations with selection based on arg max log-likelihood, or structural parameter constraints (e.g., monotonicity constraints such as bin 1 $\geq$ bin 2 $\geq \ldots$).
- Evaluate whether total variation relates or not to forecasts. Is the variability of the forecasts more dominated by the sampling error vs ... Does the deconvolution improve forecasts relative to the convolution? In which way ? bias , variability?
- Why does the em-alg behave differently for each model? Is decon-

6 Appendix

6.1 Simulation Parameters

Table 2: Parameters used to construct the Leslie matrix.

Age	Survival	p_coy	Litter_size	F_i
1	0.50	0.00	0.00	0.00
2	0.89	0.00	0.00	0.00
3	0.95	0.00	0.00	0.00
4	0.99	0.00	0.00	0.00
5	0.98	0.15	2.12	0.31
6	0.96	0.35	2.25	0.76
7	0.96	0.55	2.40	1.27
8	0.96	0.55	2.40	1.27
9	0.96	0.55	2.40	1.27
10	0.96	0.55	2.40	1.27
11	0.96	0.55	2.40	1.27
12	0.96	0.70	2.60	1.75
13	0.96	0.70	2.60	1.75
14	0.96	0.70	2.60	1.75
15	0.96	0.70	2.60	1.75
16	0.96	0.70	2.60	1.75
17	0.96	0.58	2.75	1.53
18	0.96	0.58	2.75	1.53
19	0.96	0.58	2.75	1.53
20	0.96	0.58	2.75	1.53

6.2 Results

model	sample_size	mean	median	upper_95
True distribution	10	0.412	0.410	0.527
True distribution	25	0.280	0.280	0.367
True distribution	50	0.198	0.200	0.271
True distribution	100	0.140	0.135	0.185
True distribution	150	0.113	0.113	0.148
True distribution	200	0.102	0.099	0.131

Table 3: Summary of total variation distance for samples from true distribution, based on 100 replications per sample size. Total variation distance is computed relative to the true distribution.

model	sample_size	mean	median	upper_95
A	10	0.433	0.433	0.585
A	25	0.308	0.302	0.414
A	50	0.249	0.250	0.334
A	100	0.199	0.194	0.266
A	150	0.175	0.174	0.230
A	200	0.166	0.163	0.218
B	10	0.528	0.517	0.682
B	25	0.401	0.395	0.514
B	50	0.331	0.336	0.403
B	100	0.286	0.282	0.340
B	150	0.265	0.267	0.315
B	200	0.261	0.261	0.306
C	10	0.675	0.668	0.788
C	25	0.598	0.600	0.678
C	50	0.553	0.555	0.627
C	100	0.519	0.518	0.564
C	150	0.506	0.505	0.548
C	200	0.499	0.493	0.543
D	10	0.580	0.591	0.711
D	25	0.456	0.451	0.558
D	50	0.422	0.422	0.498
D	100	0.380	0.379	0.434
D	150	0.373	0.374	0.424
D	200	0.362	0.364	0.407

Table 4: Summary of total variation distance statistics from sample simulations across convoluted distributions and different sample sizes, based on 100 replicas per sample siza. Total variation distance is computed relative to the true distribution.

model	sample_size	mean	median	upper_95
A	10	0.554	0.539	0.757
A	25	0.465	0.449	0.630
A	50	0.396	0.381	0.587
A	100	0.332	0.314	0.495
A	150	0.282	0.276	0.410
A	200	0.265	0.247	0.403
B	10	0.516	0.532	0.734
B	25	0.392	0.382	0.625
B	50	0.316	0.307	0.503
B	100	0.246	0.240	0.406
B	150	0.218	0.216	0.354
B	200	0.202	0.190	0.339
C	10	0.613	0.588	0.819
C	25	0.507	0.493	0.669
C	50	0.435	0.428	0.579
C	100	0.387	0.383	0.522
C	150	0.343	0.333	0.447
C	200	0.336	0.329	0.421
D	10	0.572	0.570	0.803
D	25	0.489	0.481	0.693
D	50	0.396	0.374	0.595
D	100	0.328	0.313	0.471
D	150	0.309	0.284	0.454
D	200	0.292	0.284	0.392

Table 5: Summary of total variation distance statistics from sample simulations across uniformed deconvolutions and different sample sizes, based on 100 replicas per sample siza. Total variation distance is computed relative to the true distribution.

model	sample_size	mean	median	upper_95
A	10	0.569	0.547	0.779
A	25	0.466	0.445	0.658
A	50	0.386	0.372	0.578
A	100	0.325	0.323	0.449
A	150	0.274	0.264	0.396
A	200	0.249	0.242	0.369
B	10	0.533	0.550	0.745
B	25	0.423	0.439	0.681
B	50	0.345	0.348	0.571
B	100	0.251	0.246	0.425
B	150	0.211	0.189	0.393
B	200	0.175	0.164	0.353
C	10	0.595	0.580	0.829
C	25	0.471	0.459	0.682
C	50	0.386	0.365	0.634
C	100	0.296	0.277	0.459
C	150	0.260	0.246	0.386
C	200	0.232	0.223	0.368
D	10	0.540	0.569	0.818
D	25	0.399	0.378	0.704
D	50	0.285	0.280	0.525
D	100	0.204	0.192	0.398
D	150	0.183	0.167	0.345
D	200	0.159	0.156	0.273

Table 6: Summary of total variation distance statistics from sample simulations across informed deconvolutions and different sample sizes, based on 100 replicas per sample siza. Total variation distance is computed relative to the true distribution.